

Nine Education IIT Academy

Class 11 — Physics Test Paper

SET A

Name: _____ Roll No.: _____ Class/Section:

_____ Date: _____

Total Marks: 30

Instructions: Answer all questions. Show all working clearly.

Section 1 — 2 Marks Questions ($5 \times 2 = 10$ Marks)

1. Convert 150° to radian.
 2. Write the $\cos 2\theta$ formula (all 3 forms).
 3. Find the magnitude of the vector $3\hat{i} + 4\hat{j} + 5\hat{k}$.
 4. Find $\mathbf{A} \times \mathbf{B}$, where $\mathbf{A} = \hat{i} - 3\hat{j} + 4\hat{k}$ and $\mathbf{B} = 2\hat{i} + 4\hat{j} + 3\hat{k}$.
 5. If $y = x^4 + 4\cos x + e^x$, find dy/dx .
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Section 2 — 4 Marks Questions ($3 \times 4 = 12$ Marks)

1. $\mathbf{A} = 2\hat{i} + \hat{j} - \hat{k}$, $\mathbf{B} = \hat{i} - \hat{k}$. (a) Find the resultant of $\mathbf{A}$ and $\mathbf{B}$. (b) Find the angle between $\mathbf{A}$ and $\mathbf{B}$. (c) Find the projection of $\mathbf{A}$ on $\mathbf{B}$.
2. If $y = \sin^5(x + 2)$, find dy/dx and d^2y/dx^2 .
3. Van der Waals' equation is $(P - a/V^2)(V - b) = nRT$, where P is the pressure of the gas, V is the volume of the gas, and T is the temperature of the gas. Find the

dimensions of a and b.

Section 3 — 8 Marks Question ($1 \times 8 = 8$ Marks)

1. A stone is dropped from the top of a tower of height h and reaches the ground after time t . Derive expressions for t and the striking velocity in terms of h and g . Hence find the time taken and the velocity with which the stone strikes the ground when dropped from a tower 122.5 m high. ($g = 9.8 \text{ m/s}^2$)
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All the best!